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2 Real Analysis 2023
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Test 2
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Instructions:

- This question paper consists of **9 pages** (including this one).
- You have 90 minutes to answer the questions.
- Answer all questions in the corresponding frames.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 50 points = 100% (52 points available).

Student Number:									
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Problem 1

Let $(x_n)_{n=1}^{\infty}$ be the sequence defined recursively by

$$x_1 = 2,$$

$$x_{n+1} = 4 - \frac{3}{x_n}, \quad n \in \mathbb{N}.$$

(a) Prove that $2 \leq x_n \leq x_{n+1} \leq 4$, $n \in \mathbb{N}$. (5 points)

Solution. We use induction in n . Notice that $x_2 = 4 - \frac{3}{x_1} = 4 - \frac{3}{2} = \frac{5}{2}$.

Induction base: We consider $n = 1$. We have $2 \leq \underbrace{2}_{=x_1} \leq \underbrace{\frac{5}{2}}_{=x_2} \leq 4$.

Induction step: Assume that $2 \leq x_m \leq x_{m+1} \leq 4$ for some $m \in \mathbb{N}$.

We have to show that $2 \leq x_{m+1} \leq x_{m+2} \leq 4$.

We start with $2 \leq x_m \leq x_{m+1} \leq 4$.

Take the reciprocal: $\frac{1}{4} \leq \frac{1}{x_{m+1}} \leq \frac{1}{x_m} \leq \frac{1}{2}$.

Multiply by -3 : $-\frac{3}{2} \leq -\frac{3}{x_m} \leq -\frac{3}{x_{m+1}} \leq -\frac{3}{4}$.

Add 4: $4 - \frac{3}{2} \leq 4 - \frac{3}{x_m} \leq 4 - \frac{3}{x_{m+1}} \leq 4 - \frac{3}{4}$.

We obtain $2 < \frac{5}{2} \leq x_{m+1} \leq x_{m+2} \leq \frac{13}{4} < 4$.

The induction step follows.

By the Principle of Mathematical Induction we conclude that $2 \leq x_n \leq x_{n+1} \leq 4$ for all $n \in \mathbb{N}$.

Student Number:								
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(b) Formulate the Monotone Convergence Theorem. (2 points)

Solution. If $(x_n)_{n=1}^{\infty}$ is a sequence that is increasing and bounded above, then it converges (to its supremum).

(c) Explain how the Monotone Convergence Theorem is applied to show that the sequence $(x_n)_{n=1}^{\infty}$ converges. (2 points)

Solution. We have shown in (a) that $x_n \leq x_{n+1}$ for all $n \in \mathbb{N}$; this means that $(x_n)_{n=1}^{\infty}$ is increasing. Moreover, we have shown that $x_n \leq 4$ for all $n \in \mathbb{N}$; this means that $(x_n)_{n=1}^{\infty}$ is bounded above. By the Monotone Convergence Theorem we conclude that $(x_n)_{n=1}^{\infty}$ converges.

(d) Find $\lim_{n \rightarrow \infty} x_n$. (3 points)

Solution. Let $x = \lim_{n \rightarrow \infty} x_n$.

Taking the limit as $n \rightarrow \infty$ in the recurrence $x_{n+1} = 4 - \frac{3}{x_n}$, we obtain that

$$x = 4 - \frac{3}{x}.$$

This equation is equivalent to the quadratic equation

$$x^2 - 4x + 3 = 0.$$

The solutions are $x = 2 \pm \sqrt{4 - 3} = 2 \pm 1$, so that $x = 1$ or $x = 3$. Recall that $x_n \geq 2$ for all $n \in \mathbb{N}$. By the Order Limit Theorem this implies that $x = \lim_{n \rightarrow \infty} x_n \geq 2$.

We conclude that $\lim_{n \rightarrow \infty} x_n = 3$.

Student Number:								
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Problem 2

(a) Consider the series $\sum_{n=1}^{\infty} \frac{2}{(2n-1)(2n+1)}$. Find an expression for its partial sums. Use this expression to show that the series converges, and find its sum. (3 points)

Solution. We observe that

$$\frac{2}{(2n-1)(2n+1)} = \frac{1}{2n-1} - \frac{1}{2n+1}.$$

Therefore, our series is a telescopic series. For the partial sums s_N with $N \in \mathbb{N}$ we obtain

$$\begin{aligned} s_N &= \sum_{n=1}^N \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \\ &= \left(1 - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \cdots + \left(\frac{1}{2N-1} - \frac{1}{2N+1} \right) \\ &= 1 - \frac{1}{2N+1}. \end{aligned}$$

As $N \rightarrow \infty$ we obtain $\lim_{N \rightarrow \infty} s_N = \lim_{N \rightarrow \infty} \left(1 - \frac{1}{2N+1} \right) = 1$.

It follows that the series converges and $\sum_{n=1}^{\infty} \frac{2}{(2n-1)(2n+1)} = 1$.

Student Number:									
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(b) For each of the following series, determine whether it is convergent or divergent. Give a justification.

(i) $\sum_{n=0}^{\infty} \frac{n!(n+1)!}{(2n)!}$ (2 points)

Solution. We apply the Ratio Test:

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!(n+2)!}{(2n+2)!} \cdot \frac{(2n)!}{n!(n+1)!} = \lim_{n \rightarrow \infty} \frac{(n+2)(n+1)}{(2n+2)(2n+1)} \\ &= \lim_{n \rightarrow \infty} \frac{n+2}{2(2n+1)} = \lim_{n \rightarrow \infty} \frac{1 + \frac{2}{n}}{2\left(2 + \frac{1}{n}\right)} = \frac{1}{4} < 1. \end{aligned}$$

By the Ratio Test, the series converges.

(ii) $\sum_{n=1}^{\infty} \left(\frac{3n^2 + 2n - 2}{2n^2 + 5n} \right)^n$ (2 points)

Solution. We apply the Root Test:

$$L = \lim_{n \rightarrow \infty} |a_n|^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{3n^2 + 2n - 2}{2n^2 + 5n} = \lim_{n \rightarrow \infty} \frac{3 + \frac{2}{n} - \frac{2}{n^2}}{2 + \frac{5}{n}} = \frac{3}{2} > 1.$$

By the Root Test, the series diverges.

(iii) $\sum_{n=1}^{\infty} \frac{5 + (-1)^n}{n\sqrt{n}}$ (2 points)

Solution. We use the Comparison Test. For any $n \in \mathbb{N}$ we have

$$0 < \frac{5 + (-1)^n}{n\sqrt{n}} \leq \frac{6}{n^{\frac{3}{2}}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{3}{2}}}$ is a convergent p -series with $p = \frac{3}{2}$. Therefore also the series $\sum_{n=1}^{\infty} \frac{6}{n^{\frac{3}{2}}}$ converges. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{5+(-1)^n}{n\sqrt{n}}$ converges.

Student Number:								
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(c) For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent. Give a justification.

(i) $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^2+1}}$ (4 points)

Solution.

First we consider the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^2+1}}$. For any $n \in \mathbb{N}$ we have

$$\frac{1}{\sqrt{n^2+1}} \geq \frac{1}{\sqrt{n^2+n^2}} = \frac{1}{\sqrt{2}n}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n}$ is the divergent harmonic series, hence also the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2}n}$ diverges.

By the Comparison Test, the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^2+1}}$ diverges.

For the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^2+1}}$ it means that it does not converge absolutely.

The series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^2+1}}$ is an alternating series. We have $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+1}} = 0$,

and $\frac{1}{\sqrt{n^2+1}} \geq \frac{1}{\sqrt{(n+1)^2+1}}$ for all $n \in \mathbb{N} \cup \{0\}$. By the Alternating Series Test, the series

$\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^2+1}}$ converges.

We conclude that the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^2+1}}$ converges conditionally.

(ii) $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^4+1}}$ (3 points)

Solution.

We consider the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^4+1}}$. For any $n \in \mathbb{N}$ we have

$$0 < \frac{1}{\sqrt{n^4+1}} \leq \frac{1}{\sqrt{n^4}} = \frac{1}{n^2}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges. By the Comparison Test, the series $\sum_{n=0}^{\infty} \frac{1}{\sqrt{n^4+1}}$ converges.

We conclude that the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{\sqrt{n^4+1}}$ converges absolutely.

Student Number:									
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Problem 3

Read the following theorem and an outline of its proof.

Theorem. The intersection of a finite collection of open sets is open.

Proof. Let $\{G_1, G_2, \dots, G_n\}$ be a finite collection of open sets. If $\bigcap_{i=1}^n G_i = \emptyset$, then we are done, so assume that $\bigcap_{i=1}^n G_i \neq \emptyset$. Take $x \in \bigcap_{i=1}^n G_i$. Then $x \in G_j$ for all $j = 1, 2, \dots, n$. For each $j = 1, 2, \dots, n$, there exists $\varepsilon_j > 0$ such that $V_{\varepsilon_j}(x) \subseteq G_j$. It follows that there is $\varepsilon > 0$ that $V_\varepsilon(x) \subseteq \bigcap_{i=1}^n G_i$. Thus, $\bigcap_{i=1}^n G_i$ is an open set.

Answer the following questions:

(a) What is $V_\varepsilon(x)$? (1 points)

Solution. For $\varepsilon > 0$ we define $V_\varepsilon(x) = (x - \varepsilon, x + \varepsilon)$.

(b) Give a definition of open set. (2 points)

Solution. A set G is an open set if for any $x \in G$ there exists $\varepsilon > 0$ such that $V_\varepsilon(x) \subseteq G$.

(c) Why do we claim that we are done if $\bigcap_{i=1}^n G_i = \emptyset$? (1 points)

Solution. Because the empty set $\emptyset$ is open.

(d) Explain why there exists $\varepsilon_j > 0$ such that $V_{\varepsilon_j}(x) \subseteq G_j$. (1 points)

Solution. This follows from the fact that each G_j is open by the definition of open set.

(e) How do we choose $\varepsilon > 0$? (2 points)

Solution. We take $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\} > 0$.

(f) Explain why $V_\varepsilon(x) \subseteq \bigcap_{i=1}^n G_i$. (2 points)

Solution. Let $j \in \{1, \dots, n\}$. Since $\varepsilon \leq \varepsilon_j$, we have

$$V_\varepsilon(x) \subseteq V_{\varepsilon_j}(x) \subseteq G_j.$$

Since this is true for any $j \in \{1, \dots, n\}$, it follows that $V_\varepsilon(x) \subseteq \bigcap_{i=1}^n G_i$.

(g) Give an example that shows that the intersection of countably many open sets need not be open. (2 points)

Solution. Put, for example, $G_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$. Then $\bigcap_{n=1}^\infty G_n = \{0\}$ which is not an open set.

Student Number:								
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Problem 4

Consider the set $S = \left\{-\frac{1}{n} : n \in \mathbb{N}\right\} \cup [0, 2] \cup \{3\}$.

(a) Determine, without proof, the set S' of all limit points of S . (1 points)

Solution. $S' = [0, 2]$.

(b) Determine, without proof, the set of all isolated points of S . (1 points)

Solution. The set of isolated points of S is $\left\{-\frac{1}{n} : n \in \mathbb{N}\right\} \cup \{3\}$.

(c) Is S compact? Justify your answer. (3 points)

Solution. A subset of $\mathbb{R}$ is compact if and only if it is closed and bounded.

The closure of S is $\overline{S} = S \cup S' = S$. It follows that S is closed.

Moreover, S is bounded: for any $x \in S$ we have $-1 \leq x \leq 3$.

We conclude that S is compact.

Student Number:									
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Problem 5

(a) Let $f : A \rightarrow \mathbb{R}$ be a function, let a be a limit point of A and $L \in \mathbb{R}$. Define what it means that $\lim_{x \rightarrow a} f(x) = L$. (2 points)

Solution. By definition we say that $\lim_{x \rightarrow a} f(x) = L$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall x \in A : 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

In other word, for any $\varepsilon > 0$ there exists $\delta > 0$ such that $|f(x) - L| < \varepsilon$ for all $x \in A$ with $0 < |x - a| < \delta$.

(b) Let $f : \mathbb{R} \rightarrow \mathbb{R}$. Suppose that $\lim_{x \rightarrow 1} f(x) = 2$. Prove by definition that

$$\lim_{x \rightarrow 1} \frac{(f(x))^2 + 2}{2} = 3.$$

(6 points)

Solution. We have

$$\left| \frac{(f(x))^2 + 2}{2} - 3 \right| = \left| \frac{(f(x))^2 + 2 - 6}{2} \right| = \frac{1}{2} |(f(x))^2 - 4| = \frac{1}{2} |f(x) - 2| |f(x) + 2|.$$

By the definition of the limit $\lim_{x \rightarrow 1} f(x) = 2$ there exists $\delta_1 > 0$ such that

$$0 < |x - 1| < \delta_1 \implies |f(x) - 2| < 1.$$

For such x we have $-1 < f(x) - 2 < 1$, and $1 < f(x) < 3$. We conclude that $3 < f(x) + 2 < 5$, $|f(x) + 2| < 5$.

Thus, for $0 < |x - 1| < \delta_1$ we have

$$\left| \frac{(f(x))^2 + 2}{2} - 3 \right| = \frac{1}{2} |f(x) - 2| |f(x) + 2| < \frac{5}{2} |f(x) - 2|.$$

Given $\varepsilon > 0$, there is $\delta_2 > 0$ such that

$$0 < |x - 1| < \delta_2 \implies |f(x) - 2| < \frac{\varepsilon}{5}.$$

Put $\delta = \min\{\delta_1, \delta_2\} > 0$. We have

$$0 < |x - 1| < \delta_2 \implies \left| \frac{(f(x))^2 + 2}{2} - 3 \right| < \frac{5}{2} |f(x) - 2| < \frac{5}{2} \cdot \frac{\varepsilon}{5} = \varepsilon.$$

By definition, $\lim_{x \rightarrow 1} \frac{(f(x))^2 + 2}{2} = 3$.